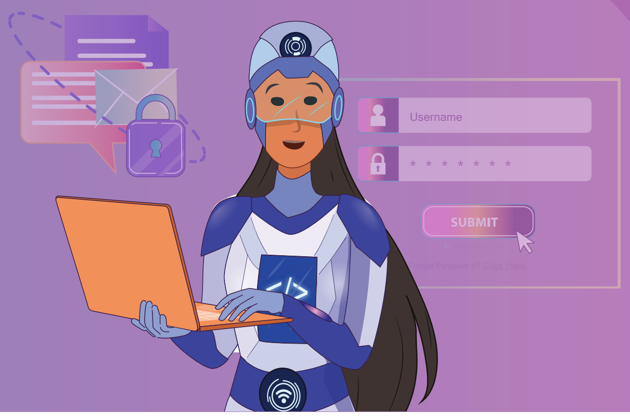


Cyber Security

12

The internet is a common tool in our daily lives. We all use it every day to pay bills, stream shows, watch videos, send emails, play games, etc. All the data on the internet is shared with billions of people. So, it is very important to keep the data secure.

Recall:

Computer Network Structure

- *Our personal computers are connected to the router.*
- *The router is connected to the ISP (Internet Services Provider).*
- *Since we are on the internet, we might be connected to a service like an app or use a search engine. The computer is then linked with the respective domain name server.*
- *The server provides the information that we request. The data then travels back to our computer.*

Why do we need to secure our data?

Would you be comfortable living in a house that other people have a key to? Or if your windows or doors never close all the way? You wouldn't feel safe in a house like that.

The internet is similar to a house like this. When we use the internet, we expose ourselves to the world. Just like real-world thieves, some data thieves could be able to steal our information and misuse it.

For example: If a banking ID and password are leaked, a hacker can easily take over the account and steal the hard-earned money in it.

Cybersecurity

Cybersecurity is the practice of protecting a network, system, or program from data leaks. In this process, we protect our data from malicious threats over the internet. This includes defending our personal and confidential information from virtual thieves.

The word 'Cyber' stands for anything related to computers or computer networks.

Security means to protect or guard.

Types of cybersecurity threats

There are different types of cyber threats. An average user is most likely to face the following kinds:

a. Viruses and Malware

b. Phishing

c. Password Attack

d. Man in the Middle Attack (MITM)

Computer Viruses

A computer virus is a program that can copy and modify its code in the system. It then corrupts data in the system by either erasing it, destroying it, or taking away the user's access to it.

Malware

Malware, or malicious software, is any software or block of code that damages a computer network, client, or server. It includes all types of attacks that can be carried out by software or code. A virus is a type of malware.

Other examples of malware are -

a. Ransomware: Locks the user's access to data in exchange for money.

b. Spyware: Spies on data that travels from the user's computer.

How can a virus or malware infect a computer?

a. Pop up ads on the internet.

b. Downloading infected software.

c. Opening unknown email attachments.

d. Inserting or connecting to a virus-infected disk or drive.

e. Visiting unknown links.

Phishing

Phishing is a cybercrime where an attacker sends emails that appear to be from a trusted source. These are used to extract the personal information of the user. This is often done via social engineering and trickery.

Social engineering comprises various methods to find information about a person like their name, email, phone number, the bank they use, the car they drive, their pending bills, etc.

An example of phishing is:

a. The user receives an email from a credit card company to pay their bills.

b. The user checks the website, and it looks authentic.

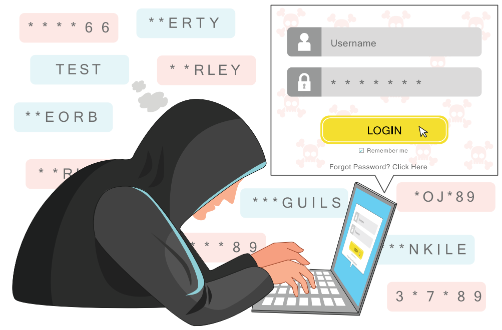
c. The user is asked to fill a form and send their information, or to make a payment online.

d. The attacker then notes the user's credentials.

Password attack

We use passwords to gain access to our accounts. A password attack is done by looking around a user's notes or guessing common passwords.

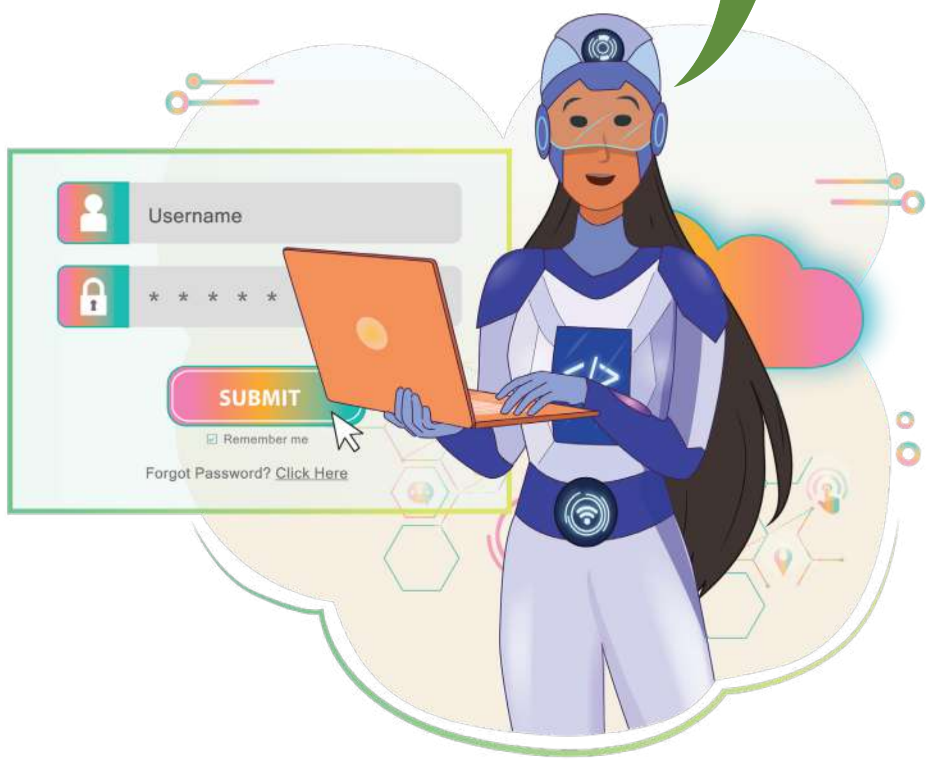
A weak password, which can be guessed easily, is prone to password attacks. This means that it can eventually be guessed through multiple attempts.



Bytes

There is no way to predict a phishing attack. Phishing emails are mostly filtered and sent to the spam. The attack is not successful until we open the link in the email or provide our details.

We need to follow some basic rules while setting a password. These will safeguard us from password attacks.



a. Use alphabets, numbers, and special characters in the password.

b. Never use something obvious like name, email ID, or birthdate as a password.

c. The password must be at least 8 characters long.

d. Never use something predictable like a sequence of numbers or the word 'password'. For example, 1234, 0000, password1, etc.

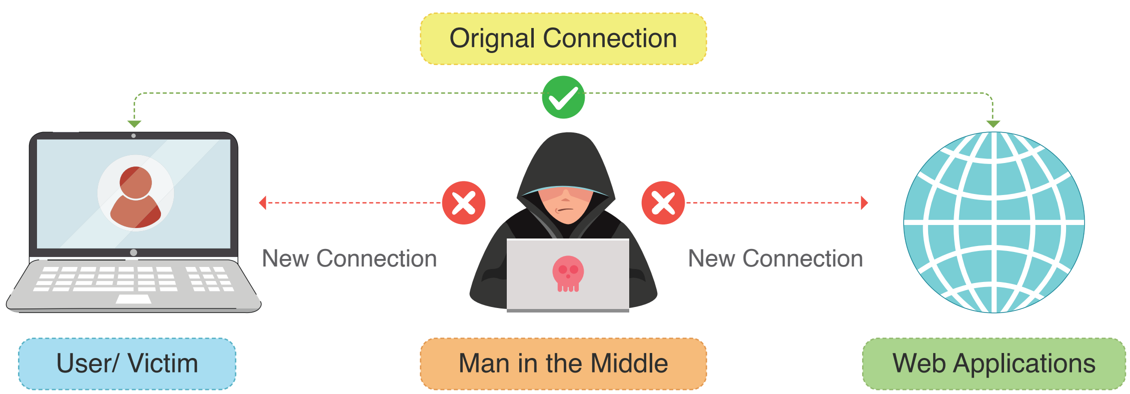
e. Use tools like Random Password Generator to create a password if required.

f. Do not use the same password for different accounts.

Man in the Middle Attack (MITM)

MITM refers to a situation where the attacker positions itself between the user and the server. Once successful, the attacker can scan the data that has been sent or received by the user.

A passive attack is the most common way to do this, in which an attacker makes an open public Wi-Fi hotspot. Once a user connects to this free hotspot, the attacker gains full access to any online data exchange.



Exercise

Fill in the blanks

1. The practice of protecting a network, system, or program from data leaks is called __________________.

2. __________________corrupts or erases computer data in a system.

3. ____________________, _________________ and ________________ are types of computer viruses.

Activity 1

- Make a list of 10 different passwords.

Visit - **haveibeenpwned.com/Passwords** to check if the passwords are secure.

Generate 10 random passwords using this tool - **www.random.org/passwords/**

To protect our system from these security threats, we use the following:

a. Firewall

A firewall is a security system that keeps track of and filters the incoming or outgoing data through the network. It decides which data is allowed inside the system based on a set of rules.

When we connect to the internet, we are exposed to the threat of malware and viruses. A firewall blocks such data packets from reaching the user.

A firewall system can be either hardware or software.

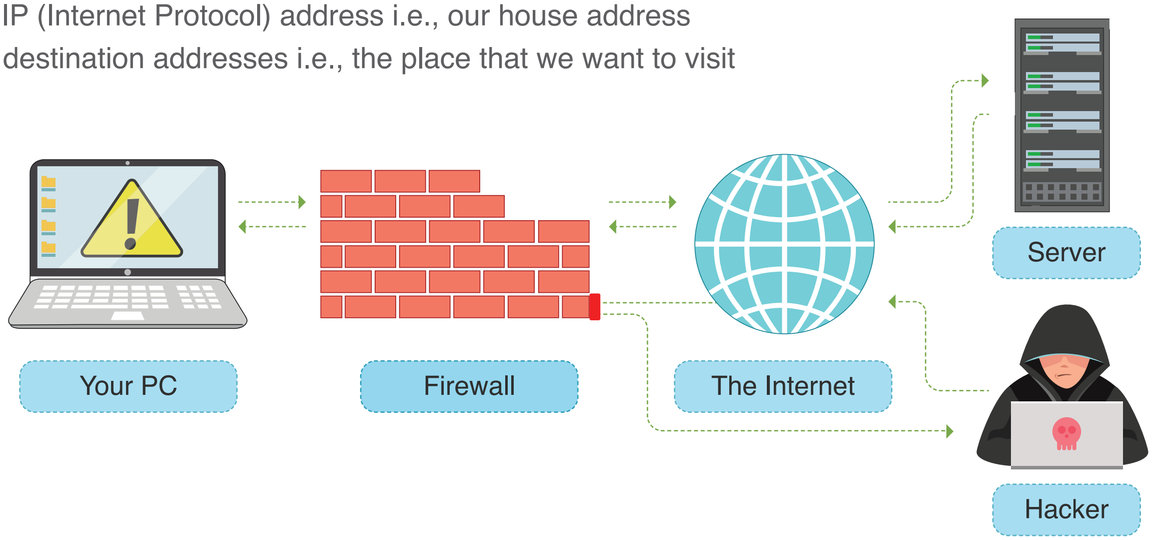
- Hardware firewalls are built-in routers. They protect devices that are connected to the network.
- Software firewalls are installed in the device itself to monitor traffic. Firewalls come preinstalled with Windows.

How does a firewall help protect data?

The computer communicates with the internet similar to how we travel outside our homes, or other people can find our homes and visit them. Information travels in the form of small bundles of data called packets.

Let us assume we are a data packet in this scenario. The information we would carry about is -

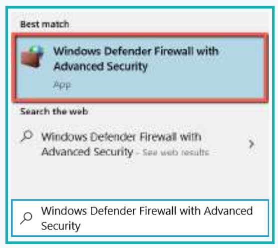
- Our IP (Internet Protocol) address i.e., our house address
- The destination addresses i.e., the place that we want to visit



Activity 2

Check the status of your network protection.

We can access Windows Defender and check the status of our network protection by typing "Windows Defender Firewall" in the search box.



b. Antiviruses

A popular method to upgrade computer security is to install a reliable anti-virus. An anti-virus is a software developed to detect and destroy computer viruses.

Sometimes, malware developers find a way to bypass the firewall. In this case, anti-viruses prove to be a wise choice to secure data. Anti-viruses have full control over managing network traffic.

Note:

Ethical Hacking is a legal practice that experts use to find flaws in the security of a system. It helps us understand network weakness and protect ourselves from upcoming threats.

Exercise

Fill in the blanks

4. ________________ is a software developed to detect and destroy computer viruses.

5. A ________________ is a security system that keeps track and filters incoming or outgoing data through the network.

Match the following:

6.

a. A legal practice to find faults in the security of a system	**g.** Man in the Middle attack
b. Safe to visit website	**h.** Strong Password
c. Default firewall installed in Windows	**i.** Antivirus
d. Attacker positions itself between the user and the server	**j.** Ethical Hacking
e. Uses alphabets, numbers, and special characters	**k.** HTTPS
f. Software that scans and removes malware	**l.** Windows Defender

While using the internet, we must be aware of security threats. Firewalls and antiviruses can help us keep our data secure. We should always take extra precautions with our data on the internet.

Tech Flash

- **Cyber Security** : The protection of internet-connected systems from cyberthreats is called cybersecurity. Cyber means anything related to computers or computer networks. Security means to protect or to guard.
- **Virus** : A computer virus is a piece of code that can copy itself. It modifies the code in the system. Then it corrupts the data in the system by either erasing it, destroying it, or taking away user's access from the data.
- **Malware** : Malware, or malicious software, is any software or block of code that is meant to intentionally damage a computer network, client, or a server.
- **Ransomeware** : Locks the user's access to data in exchange of money.
- **Spyware** : Spyware is used to spy on data that travels from a user's computer.
- **Phishing** : A phishing attack is the practice of sending emails that appear to be from a trusted source to extract a user's personal information.
- **Firewall** : A firewall is a security system that keeps track of network traffic, i.e. data passing through the network.